

Bootstrap 5.0

Lab 1 – Page Creation Guide

A practical reference for building the Dashboard page from your lab example

Goal: Learn how to construct the page instead of memorising or blindly copying one finished HTML file.

Reference version: Bootstrap 5.0 documentation. The guide focuses on the exact Bootstrap components and utility classes used in the supplied Dashboard/Todos page, plus closely related classes useful in a lab.

Official Bootstrap 5.0 documentation:

[Bootstrap 5.0 Introduction](#)

Core idea: Bootstrap classes are combined. Usually the first class activates a component, the next class changes its variant, and utility classes control spacing, alignment, colour, or layout.

What this PDF contains

- Bootstrap setup and version reference
- Navbar anatomy and classes
- Container and spacing utilities
- Table anatomy and modifiers
- Button and badge classes
- List Group component
- Colour and background utilities
- A class-by-class audit of your supplied code
- A build order for reproducing the page in a lab
- Official Bootstrap 5.0 documentation links for every major area

1. Bootstrap setup

Your page needs Bootstrap CSS before Bootstrap classes can do anything. Bootstrap 5.0 provides a jsDelivr quick-start stylesheet. For this guide, keep your documentation reference at version 5.0 so the class names and examples match your lab material.

Official page: [Bootstrap 5.0 Introduction](#)

```
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- Bootstrap 5.0 CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
```

The Bootstrap 5.0 documentation lists 5.0.2 as the CDN quick-start version. Your original example used a 5.3.2 CDN, so if the goal is to match this PDF and your saved lab documentation exactly, use the 5.0 documentation/version consistently.

When JavaScript is required

Static parts such as the table, buttons, badges, colours, spacing, and basic list groups work with Bootstrap CSS. Interactive components such as a responsive navbar collapse require Bootstrap's JavaScript bundle.

Official page: [Bootstrap 5.0 Download / CDN](#)

2. Components used in your Dashboard page

Do not treat every class as a separate component. Your page is built from a small number of components and utility classes.

Area	Main component	Classes actually used	Purpose
Header	Navbar	navbar , navbar-dark, bg-dark, navbar-expand-lg, navbar-brand, navbar-nav, nav-item, nav-link, active	Top navigation and brand
Layout	Container	container-fluid	Full-width page wrapper
Spacing	Spacing utility	mt-4 , mt-5 , ms-auto	Margin and alignment
Data	Table	table , table-striped	Dashboard data
Labels	Badge	badge , bg-success, bg-warning, text-dark	Count/status labels
Actions	Button	btn , btn-danger, btn-sm	Remove action
Todos	List Group	list-group , list-group-item, list-group-item-success/info/warning/danger	Todo/status list

The classes in bold are the key starting classes. If you remember the component's base class first, the rest becomes much easier to understand.

3. Navbar – the most important copy-from-doc component

Official documentation: [Bootstrap 5.0 Navbar](#)

Your screenshot has a dark navbar with a logo on the left and links on the right. The navbar documentation is the correct source for the component structure.

Class anatomy

Class	What it does	Think of it as
navbar	Activates Bootstrap navbar styling.	Navbar base
navbar-dark	Uses the dark navbar colour scheme for readable light text.	Dark-theme navbar
bg-dark	Sets a dark background using a background utility.	Background = dark
navbar-expand-lg	Navbar expands at the large breakpoint and above.	Responsive breakpoint
container-fluid	Makes the container full width.	Use available width
navbar-brand	Styles the site/project name or logo.	Brand area
navbar-nav	Groups navigation items.	Navigation list
nav-item	Represents one navigation item.	One menu item
nav-link	Styles a navigation link.	Clickable link
active	Marks the current/selected navigation item visually.	Current page
ms-auto	Applies automatic start margin; in an LTR layout it can push the nav toward the right.	Push right

Navbar skeleton to understand

```
<nav class="navbar ...">
  <div class="container-fluid">

    <a class="navbar-brand" href="#">Brand</a>

    <ul class="navbar-nav ...">
      <li class="nav-item">
        <a class="nav-link ..." href="#">Link</a>
      </li>
    </ul>

  </div>
</nav>
```

This is deliberately a skeleton rather than your exact finished page. In a practical exam, choose the colour, breakpoint, brand text, links, and alignment yourself.

Responsive navbar note

The full responsive example in the Bootstrap 5.0 documentation also uses **navbar-toggler**, **collapse**, and **navbar-collapse**. Those are needed when the menu should collapse behind a mobile toggle. The simple navbar in your supplied page does not include that complete collapse mechanism.

Useful related documentation: [Navbar](#) · [Layout utilities](#)

4. Container and spacing utilities

Official documentation: [Containers](#) · [Spacing](#)

Container classes control the width of the content area. Spacing utilities add margin or padding without writing your own CSS.

Class	Meaning	Typical use in a lab
container	Responsive max-width container	Normal centered page content
container-fluid	100% width at all breakpoints	Dashboard spanning the screen
container-sm / md / lg / xl / xxl	100% until a specified breakpoint, then constrained	Responsive layouts

Spacing notation

Bootstrap 5.0 spacing follows a pattern based on property + side + size. Common properties are **m** for margin and **p** for padding. Common sides are **t** top, **b** bottom, **s** start, **e** end, **x** horizontal, and **y** vertical.

Example	Read it as	Meaning
mt-4	m + t + 4	margin-top, spacing level 4
mt-5	m + t + 5	larger margin-top
ms-auto	m + s + auto	automatic start margin; useful for alignment
p-3	p + all + 3	padding on all sides
px-3	p + x + 3	horizontal padding
py-2	p + y + 2	vertical padding

Example reasoning

```
<div class="container-fluid mt-4">
  ...
</div>

<h2 class="mt-5">Todos</h2>
```

Do not memorise the numeric value first. Understand the pattern: **mt** means margin-top, while the number selects Bootstrap's spacing level.

5. Table component

Official documentation: [Bootstrap 5.0 Tables](#)

Bootstrap table styling is opt-in. The important rule is: start with **table**, then add modifier classes such as **table-striped** when you need a variation.

Class	Purpose	When to use
table	Base Bootstrap table styling	Almost every Bootstrap table
table-striped	Zebra stripes on body rows	Improve row scanning
table-hover	Hover state for body rows	Interactive/scan-friendly tables
table-bordered	Adds borders	Clear cell separation
table-borderless	Removes borders	Minimal table look
table-dark	Dark table variant	Dark table design
table-sm	Reduces cell padding	Compact tables
table-responsive	Allows horizontal scrolling on narrow screens when wrapping the table	Wide tables on mobile

Table skeleton

```
<table class="table table-striped">
  <thead>
    <tr>
      <th>Column</th>
      <th>Column</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Value</td>
      <td>Value</td>
    </tr>
  </tbody>
</table>
```

Your page uses only **table + table-striped**. The rest of the table appearance comes from Bootstrap's base table CSS and normal HTML table structure.

Good lab variations to practise: **table table-hover**, **table table-bordered**, **table table-dark table-striped**, and **table table-sm**.

6. Buttons and badges

Official documentation: [Buttons](#) · [Badges](#)

Button class formula

```
btn
↓
base button styling

btn-danger
↓
danger/red variant

btn-sm
↓
small size
```

So **btn btn-danger btn-sm** is not one special class. It is three Bootstrap classes working together.

Class	Meaning
btn	Base button styling
btn-primary / secondary / success / danger / warning / info / light / dark	Contextual button variants
btn-outline-*	Outline version of a contextual button
btn-sm	Small button
btn-lg	Large button

Small practice example

```
<button class="btn btn-primary">Save</button>
<button class="btn btn-danger btn-sm">Delete</button>
<a class="btn btn-outline-success" href="#">View</a>
```

Badge class formula

```
badge
↓
small label/count

bg-success
↓
green background

text-dark
↓
dark text
```

Class	Purpose
badge	Base badge styling
bg-success	Green background
bg-warning	Yellow background
text-dark	Dark text; useful on light backgrounds
rounded-pill	Pill-shaped rounded appearance

Practice example

```
<span class="badge bg-success">Active</span>  
<span class="badge bg-warning text-dark">Pending</span>
```

7. List Group and contextual colours

Official documentation: [List Group](#) · [Background utilities](#) · [Text colours](#)

Your Todos section is a List Group. The contextual item classes provide success, information, warning, and danger appearances.

Class	Purpose	Visual meaning
list-group	Creates the list-group container	List wrapper
list-group-item	Creates one list item	Basic item
list-group-item-success	Success contextual item	Usually green
list-group-item-info	Information contextual item	Usually blue/cyan
list-group-item-warning	Warning contextual item	Usually yellow
list-group-item-danger	Danger contextual item	Usually red

List Group skeleton

```
<ul class="list-group">
<li class="list-group-item">Normal item</li>
<li class="list-group-item list-group-item-success">Success</li>
<li class="list-group-item list-group-item-warning">Warning</li>
<li class="list-group-item list-group-item-danger">Danger</li>
</ul>
```

Colour utilities

Family	Common classes	Use
Background	bg-primary, bg-secondary, bg-success, bg-danger, bg-warning, bg-info, bg-light, bg-dark	Background colour
Text	text-primary, text-secondary, text-success, text-danger, text-warning, text-info, text-light, text-dark	Text colour

Important: background classes do not automatically set text colour. For example, Bootstrap's documentation specifically notes that **bg-light** often needs **text-dark** for readable text.

8. Exact class audit of your supplied HTML

This section is the quick revision sheet for the page you showed. It lists only the Bootstrap classes that actually appear in that code.

Class	Category	Exact role in your page
navbar	Navbar	Activates navbar component
navbar-dark	Navbar	Dark navbar colour scheme
bg-dark	Background	Dark navbar background
navbar-expand-lg	Navbar	Expansion breakpoint at lg
container-fluid	Container	Full-width wrapper
navbar-brand	Navbar	LOGO styling
navbar-nav	Navbar	Navigation list
ms-auto	Spacing	Moves nav toward the right in LTR layout
nav-item	Navbar	One menu item
nav-link	Navbar	Navigation link styling
active	State	Marks Home as active
badge	Badge	Message count / status label
bg-success	Background	Green badge
bg-warning	Background	Yellow badge
text-dark	Text	Dark text on warning badge
mt-4	Spacing	Top margin around main content
table	Table	Base Bootstrap table
table-striped	Table	Alternating table rows
btn	Button	Base Bootstrap button
btn-danger	Button	Red remove button
btn-sm	Button	Small remove button
mt-5	Spacing	Large top margin before Todos
list-group	List Group	Todos container
list-group-item	List Group	One Todo item
list-group-item-success	List Group	Green Todo item
list-group-item-info	List Group	Blue/info Todo item
list-group-item-warning	List Group	Yellow Todo item
list-group-item-danger	List Group	Red Todo item

9. Lab 1 – how to build the page yourself

Use this order in the practical. The point is to understand what you are adding at each stage.

Step 1 – Start the HTML document

Create the normal HTML5 structure and set the page title.

Step 2 – Add Bootstrap CSS

Add the Bootstrap 5.0 CDN link inside `<head>`. Do this before trying Bootstrap classes.

Step 3 – Build the navbar

Start with **navbar**. Add the colour scheme, background, breakpoint, container, brand, navigation list, items, and links.

Step 4 – Align the navigation

Use **ms-auto** when you want the navigation pushed toward the end/right in a left-to-right layout.

Step 5 – Create the page container

Use **container-fluid** if the dashboard should span the viewport. Add **mt-4** if you need top spacing.

Step 6 – Add the Dashboard heading

Use a normal heading element. Bootstrap does not need to be used for every HTML element.

Step 7 – Add the table

Start with **table**, then choose **table-striped** if you want alternating rows.

Step 8 – Add action buttons

Use **btn** plus a contextual variant such as **btn-danger**. Add **btn-sm** only if the smaller size is wanted.

Step 9 – Add badges

Use **badge** plus a background utility such as **bg-success** or **bg-warning**. Add **text-dark** when needed for contrast.

Step 10 – Add Todos

Create a **list-group** and place **list-group-item** elements inside it. Add contextual item variants.

Step 11 – Add spacing

Use **mt-5** or other spacing utilities where the sections need separation.

Step 12 – Test the page

Check desktop width, narrow width, readable colours, and whether your navbar behaviour matches what the lab asks for.

Exam strategy

If the question says *create a Bootstrap navbar*, do not start by copying the whole Dashboard. Go to the Navbar documentation, identify the base structure, then customise the brand and links. Do the same for Table, Button, Badge, and List Group.

This approach is slower once, but much safer in a practical because you can build a page even when the exact sample is different.

10. Official Bootstrap 5.0 documentation index

These are the reference pages to use for this lab. All links point to Bootstrap 5.0 documentation.

Page	Official link	Why you need it
Bootstrap 5.0 Introduction	https://getbootstrap.com/docs/5.0/	Setup, CDN, starter page
Navbar	https://getbootstrap.com/docs/5.0/components/navbar/	Navbar, brand, nav links, responsive collapse, colours
Tables	https://getbootstrap.com/docs/5.0/content/tables/	Table base class, striped, hover, borders, responsive tables
Buttons	https://getbootstrap.com/docs/5.0/components/buttons/	Button variants, sizes, outline buttons
Badges	https://getbootstrap.com/docs/5.0/components/badge/	Counts, labels, contextual backgrounds, pills
List Group	https://getbootstrap.com/docs/5.0/components/list-group/	Basic lists, active/disabled items, contextual items
Containers	https://getbootstrap.com/docs/5.0/layout/containers/	container, container-fluid, responsive containers
Spacing	https://getbootstrap.com/docs/5.0/utilities/spacing/	Margin, padding, responsive spacing notation
Colors	https://getbootstrap.com/docs/5.0/utilities/colors/	Text colour utilities
Background	https://getbootstrap.com/docs/5.0/utilities/background/	Background colour utilities
Layout Utilities	https://getbootstrap.com/docs/5.0/layout/utilities/	Display, flexbox, spacing and layout helpers
Download / CDN	https://getbootstrap.com/docs/5.0/getting-started/download/	Bootstrap CDN and JS bundle information

Version note

Your original HTML uses Bootstrap 5.3.2, while this lab guide intentionally uses the Bootstrap 5.0 documentation you selected. Do not mix versions casually when studying for a lab: use the version your instructor/course material expects.

Final revision checklist

- Can I explain what **navbar** does without looking it up?
- Can I explain why **bg-dark** is used?
- Can I explain **navbar-expand-lg**?
- Can I create a table using **table** and add stripes with **table-striped**?
- Can I construct a button from **btn + variant + size**?
- Can I create a badge using **badge + bg-***?
- Can I create a List Group from **list-group + list-group-item**?
- Can I read **mt-5** and **ms-auto** using Bootstrap's spacing notation?
- Can I find the correct official Bootstrap page when I forget a class?